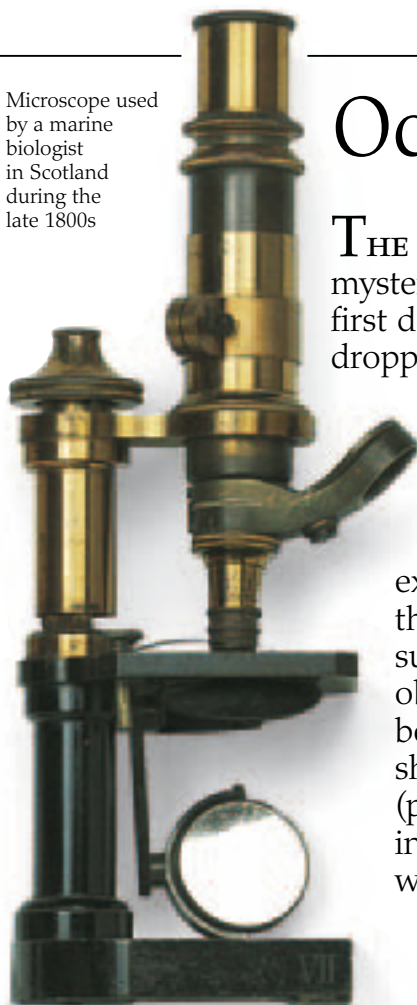


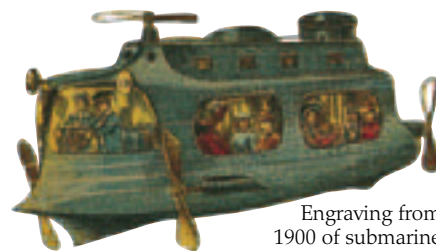
Microscope used by a marine biologist in Scotland during the late 1800s



Ocean explorers

THE OCEAN HAS ALWAYS been a place of mystery, with little to see on the surface. The first depth soundings were made by simply dropping a lead weight on a line until the

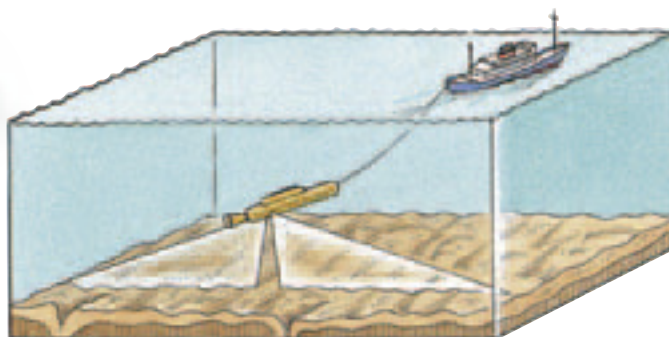
operator felt it hit the bottom. Echo sounders, invented during World War I, used single pulses of sound, which bounced back off the seabed. This led to increasingly sophisticated sonar systems, such as GLORIA. For centuries all that was known of marine life in the deep were creatures brought up in fishermen's nets or washed ashore. The HMS *Challenger* expedition of the 1870s undertook deep-sea trawls, finally showing that the deep ocean did contain marine life. The invention of manned submersibles allowed the deep-seafloor and its marine life to be directly observed. In the last 30 years startling new communities of animals have been discovered around hot springs on the ocean floor, while studies in shallow waters benefited greatly from the invention of SCUBA equipment (pp. 48–49). Today, unmanned submersibles have allowed further exploration in otherwise inaccessible waters. Yet, despite modern methods, who knows what mysteries the ocean still holds—for much of it is yet to be explored.



Engraving from 1900 of submarine bus in the year 2000

GLORIOUS GLORIA

GLORIA, for Geological Long Range Inclined Asdic (sonar), was used for over 20 years to survey the ocean floor, scanning over five percent of the world's oceans. GLORIA'S torpedo-shaped body (towfish) is 26 ft (8 m) long and weighs about 2 tons. On deck, GLORIA sat in a special cradle which was also used to launch the towfish into the water.



GLORIA AT WORK

To survey the seabed, GLORIA was towed behind its mother ship at a speed of 10 knots. Sound pulses from GLORIA would span out across the seabed up to 18 miles (30 km) on each side. GLORIA picked up echoes bouncing back from features on the seabed. These were processed by onboard computers to produce maps of the seafloor. These maps helped identify hazards on the seabed, determined routes for laying undersea cables, and assisted in exploration for valuable minerals.

